Final Project: Adversary Emulation with Atomic Red Team

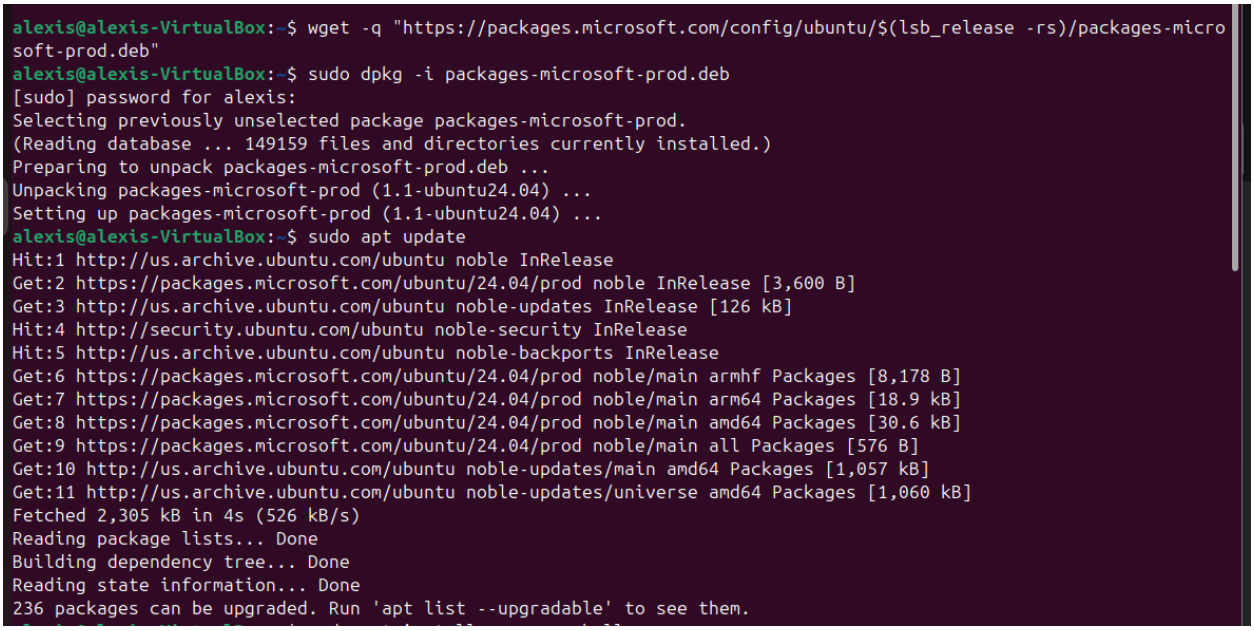
Project Members: Alexis Hernandez-Vega, Aboubakar Amadou, Sarah O'Farril-Gonzalez

Task 1 Setup and Configuration:

To begin the lab, we prepared our Linux environment by installing the necessary tools and packages required to run Atomic red Team tests through PowerShell

- Steps Completed

  Added Microsoft Package repository



This step ensured that the system could access Microsoft-hosted packages, including PowerShell.

Installed PowerShell

```
alexis@alexis-VirtualBox:~$ sudo apt install -y powershell
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  powershell
0 upgraded, 1 newly installed, 0 to remove and 236 not upgraded.
Need to get 75.5 MB of archives.
After this operation, 192 MB of additional disk space will be used.
Get:1 https://packages.microsoft.com/ubuntu/24.04/prod noble/main amd64 powershell amd64 7.5.1-1.deb [75.5 MB]
Fetched 75.5 MB in 40s (1,907 kB/s)
Selecting previously unselected package powershell.
(Reading database ... 149176 files and directories currently installed.)
Preparing to unpack .../powershell_7.5.1-1.deb_amd64.deb ...
Unpacking powershell (7.5.1-1.deb) ...
Setting up powershell (7.5.1-1.deb) ...
Processing triggers for man-db (2.12.0-4build2) ...
alexis@alexis-VirtualBox:~$ pwsh
PowerShell 7.5.1
PS /home/alexis> cd ~
```

Confirmed the installation by launching PowerShell using **pswh.**

Installed Git

Had to install Git to be able to clone the Atomic Red Team repository:

```
PS /home/alexis> sudo apt install -y git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs
  git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 236 not upgraded.
Need to get 4,804 kB of archives.
```

Imported Atomic Red Team PowerShell Module

Inside of PowerShell I imported the Atomic Red Team module

```
PS /home/alexis> Install-Module -Name invoke-atomicredteam,powershell-yaml -Scope CurrentUser -Force
PS /home/alexis> Import-Module invoke-atomicredteam                                    PS /home/alexis> $env:Atomic
```

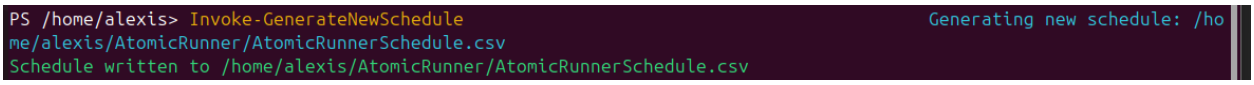
Set up Environment Variables

I configured environment variables to point to the Atomic Red Team atomics folder

```
RedTeamPath = "$HOME/AtomicRedTeam/atomics"
PS /home/alexis> $PathToAtomicsFoler = "$HOME/AtomicRedTeam/atomics"
```

Generated Atomic Test Schedule

I generated a new test schedule as well as saving the generated schedule to a specific location:



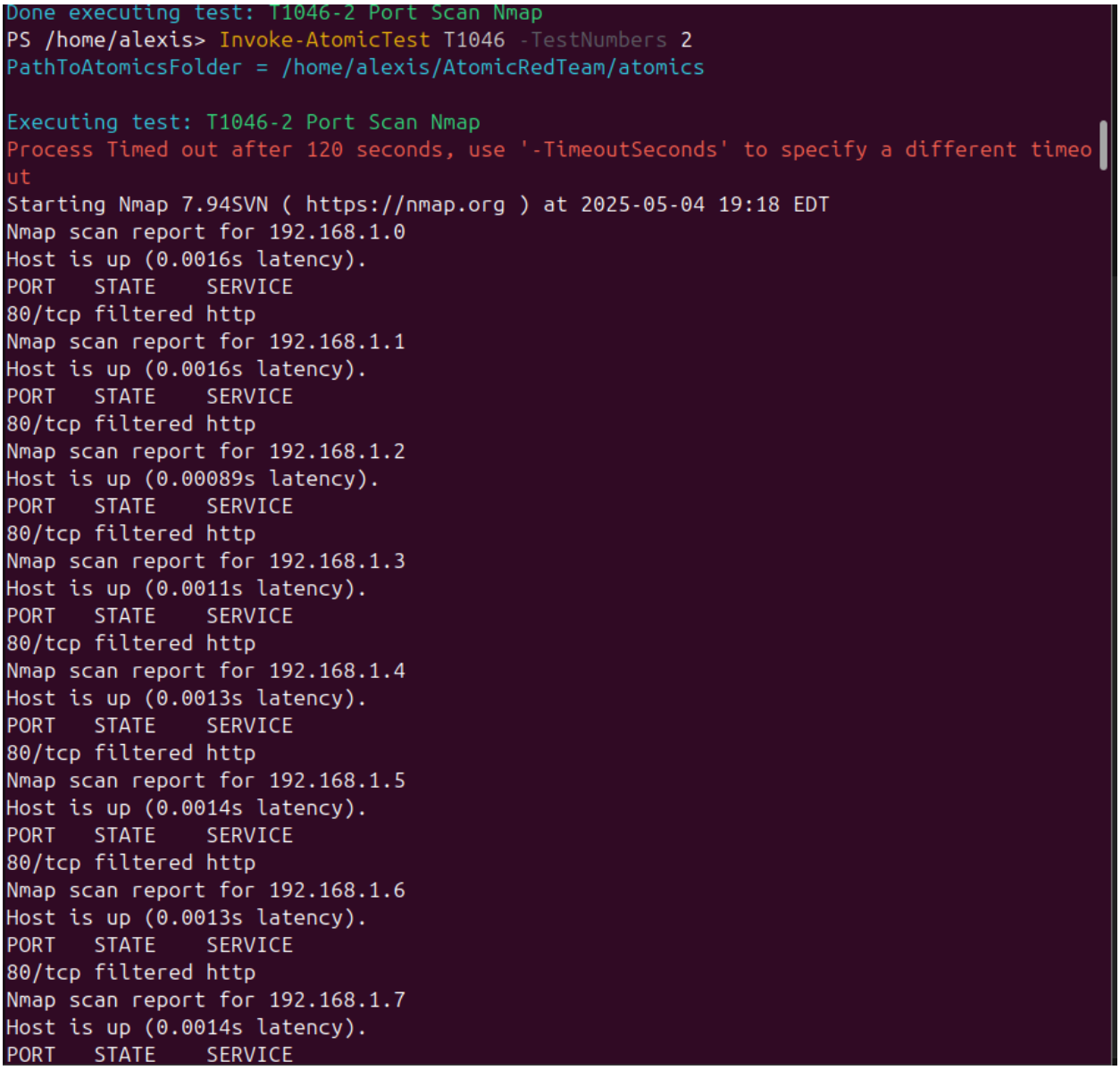
Task 2 Initial Reconnaissance

- **T1046 Network service scanning**

This technique simulates network service scanning activities, to have the ability to identify active devices and open services by sending network probes

Command Used - Invoke-AtomicTest T1046 -TestNumbers 2

This command runs Atomic Red Teams predefined Nmap port scan to simulate an attacker scanning the network.



Findings:

256 hots were detected as up in the 192.168.1.X network range.

Port 80 appeared filtered across all scanned devices.

Analysis:

Discovering 256 active hosts on a single subnet is unusual and suggest:

- A misconfiguration or simulated network with many responsive IPs
- Every IP address responding can make it easier for attackers to map and target devices

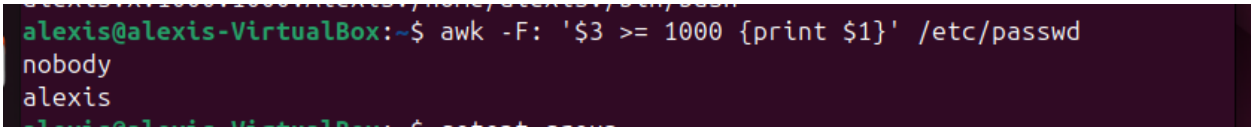
Recommendations:

- We must limit the number of accessible hots through VLANs and proper network segmentation.
- Strengthen firewall rules to minimizes the echo replies and service responses

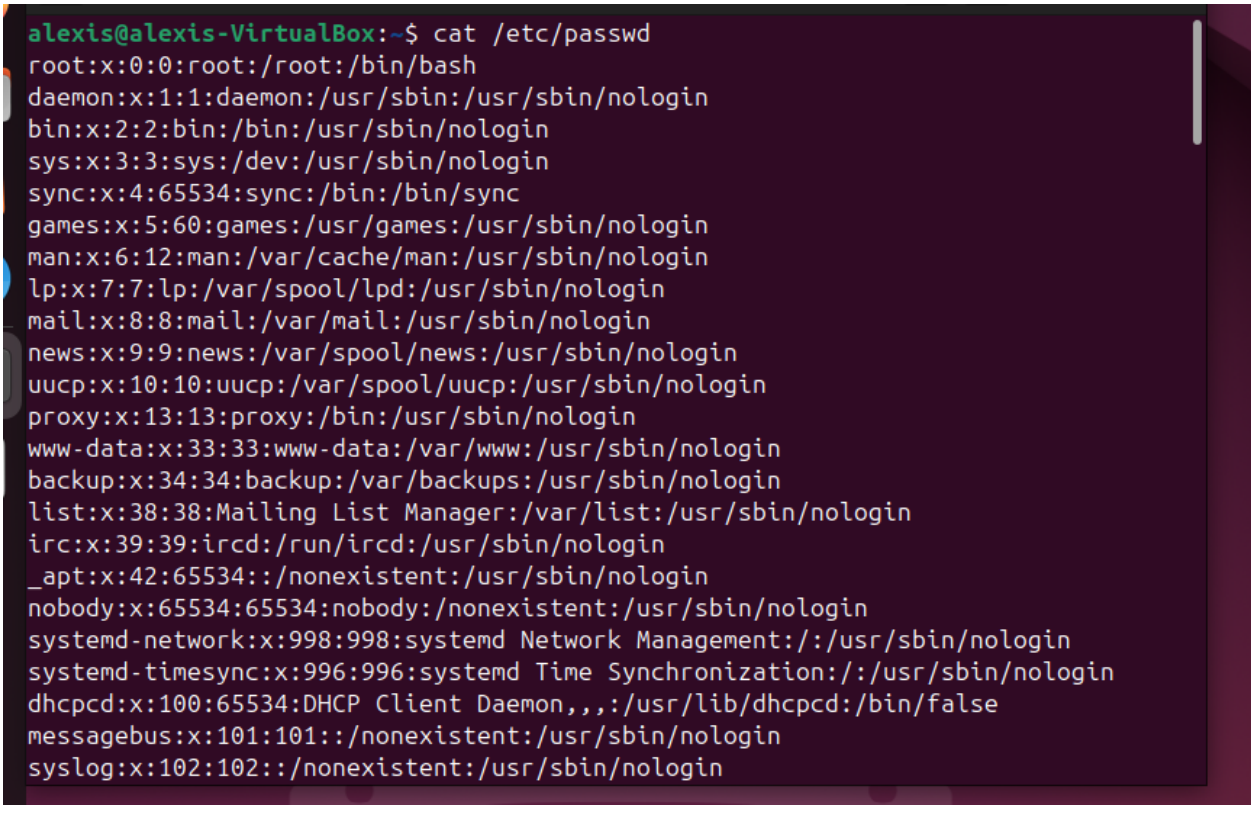
T1087 Account Discovery

Commands used:

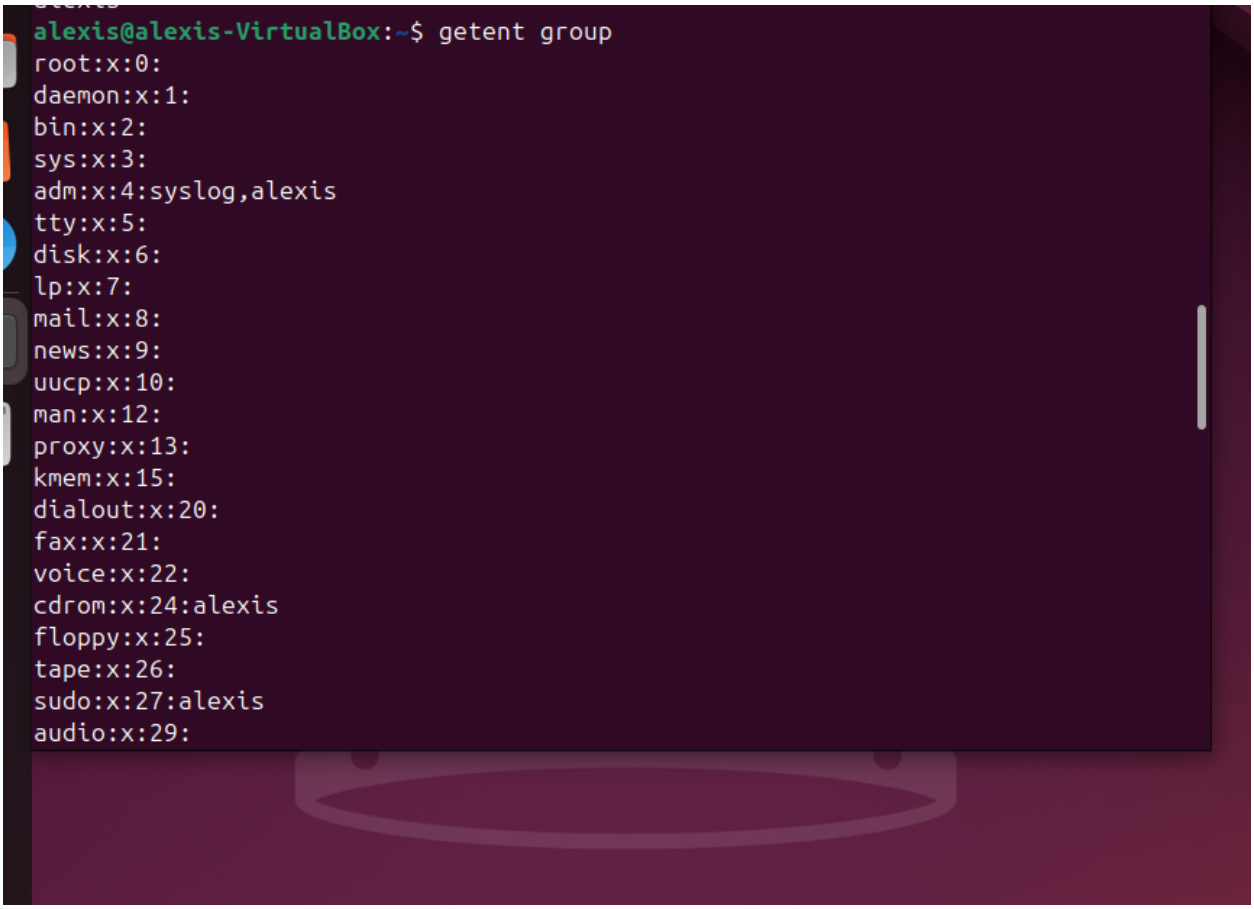
awk -F: '$3 >= 1000 {print $1}' /etc/passwd: Lists all real users by checking for user IDs greater than or equal to 1000. Real users typically have a UID of 1000 or higher



cat /etc/passwd - Displays all user accounts on the system



getent group – Listed all groups and the users assigned to them revealing important group members like sudo and adm.



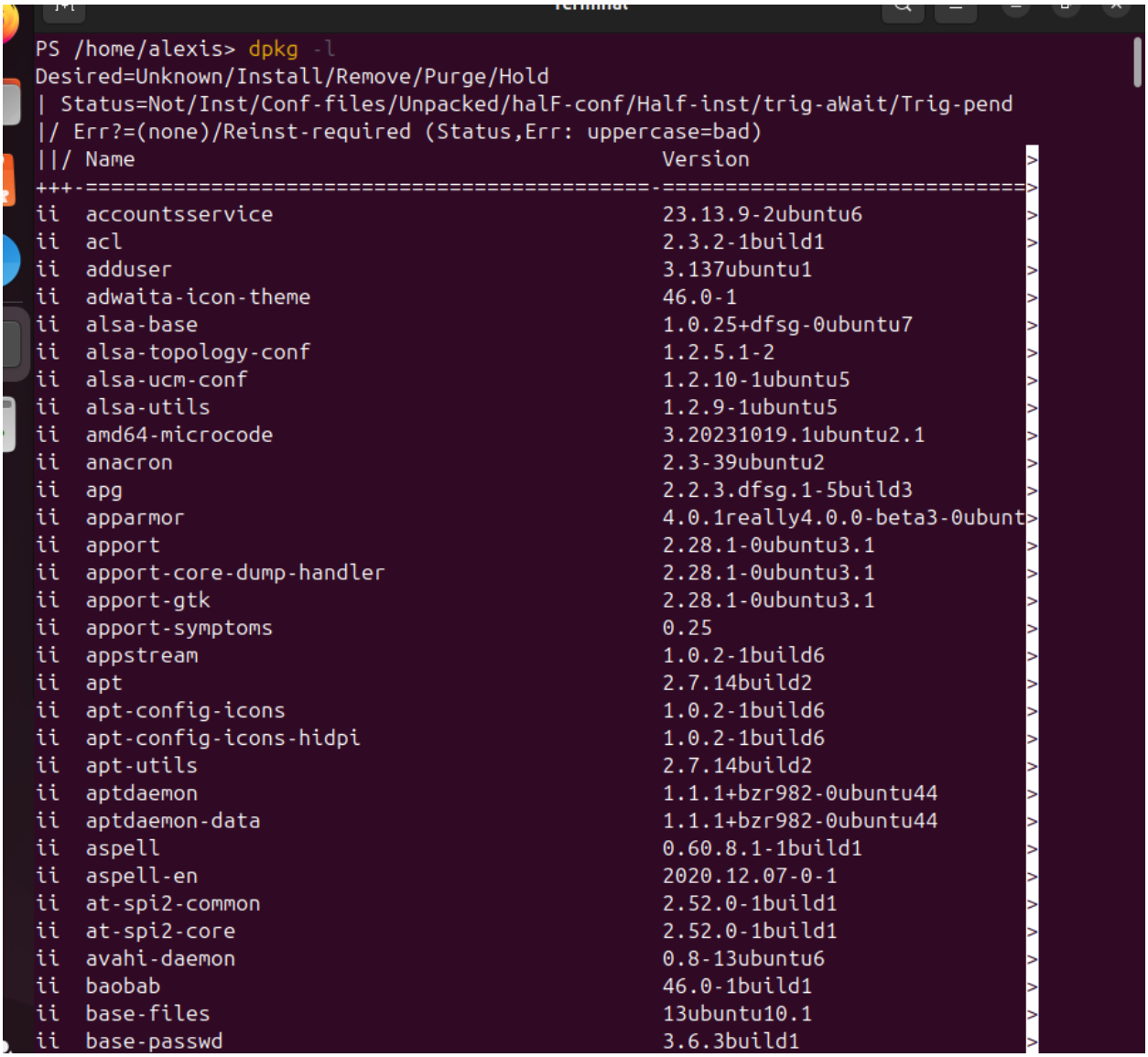
Findings:

- We were able to find the real users such as alexis and nobody
- Also able to find that alexis is a member of critical groups such as sudo,adm,cdrom,and dip granting administrative rights and elevated system access.
- In this case the alexis account is a high value target due to its Sudo privileges

Potential Vulnerabilities:

Misconfigured group memberships could increase the risk of privilege escalations if a low privilege user were added incorrectly

T1518: Software Discovery Report



The command used was dpkg –l, Software discovery is a common reconnaissance technique where an attacker attempts to enumerate installed applications and version of a target system. This information can reveal vulnerabilities or misconfigurations that can be exploited for things such as a privilege escalation or lateral movement.

Findings:

The command successfully retrieved a list of all installed software on the system:

Some examples of identified packages include:

accountsservice (version 23.13.9-2ubuntu6)

adduser (version 3.137ubuntu1)

alsa-utils (version 1.2.9-1ubuntu5)

apt (version 2.7.14build2)

The full list contained many installed applications, spanning system utilities libraries user applications and background services

Conclusion:

The software discovery process was successful. The listing of installed applications provides a view of potential outdates or vulnerable software that an attacker could exploit. Regular patching and software audits are important to minimize risks related to exposed software vulnerabilities.

Task 3 Adversary Techniques

- **T1083 -File and Directory Discovery**

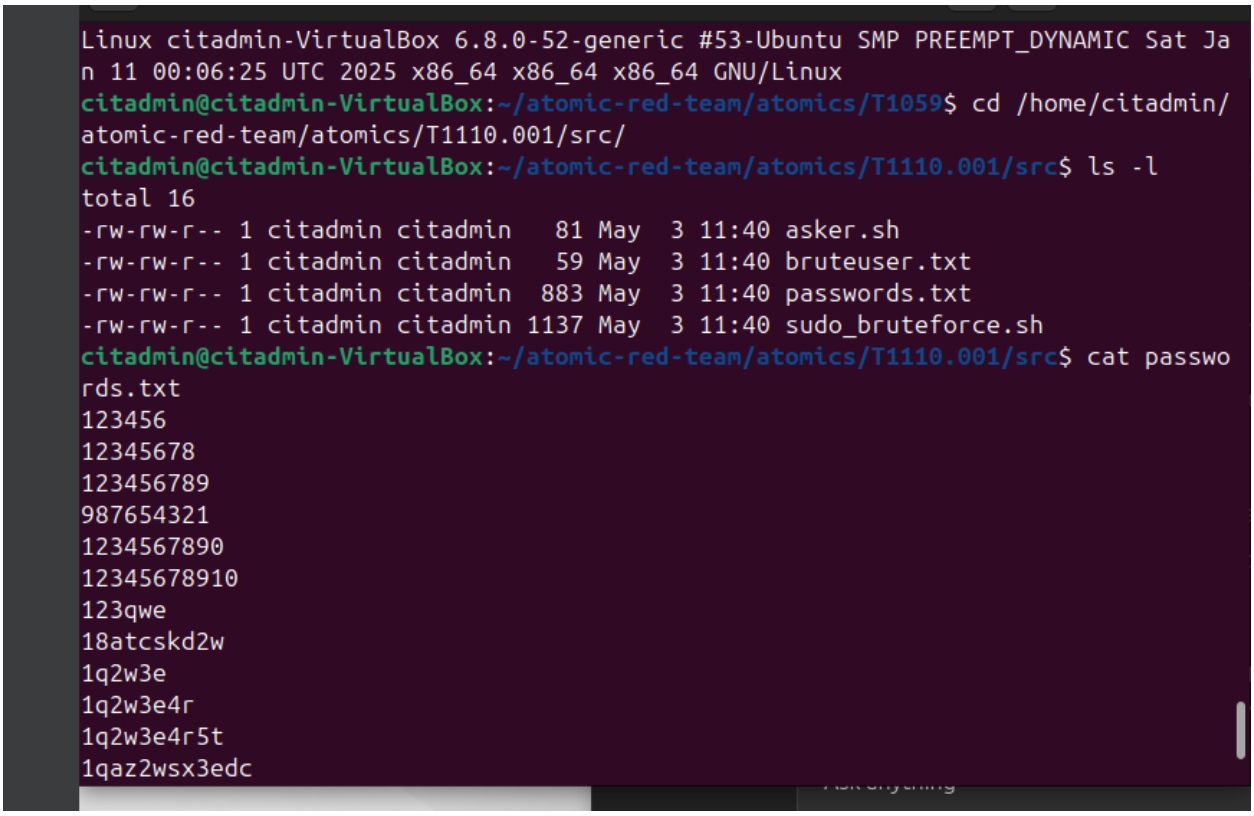
Objective: Simulate an attacker searching the file system for potentially sensitive files.

```
citadmin@citadmin-VirtualBox:~$ grep -irl "platform: linux" ~/atomic-red-team/atomics/
citadmin@citadmin-VirtualBox:~$ cd ~/atomic-red-team/atomics/T1083/
citadmin@citadmin-VirtualBox:~/atomic-red-team/atomics/T1083$ cat T1083.yaml
attack_technique: T1083
display_name: File and Directory Discovery
atomic_tests:
- name: File and Directory Discovery (cmd.exe)
  auto_generated_guid: 0e36303b-6762-4500-b003-127743b80ba6
  description: |
    Find or discover files on the file system.  Upon successful execution, this test will output the results of all the data d
iscovery commands to a specified file.
  supported_platforms:
  - windows
  input_arguments:
    output_file:
      description: File to output results to
      type: string
      default: '%temp%\T1083Test1.txt'
  executor:
    command: |
      dir /s c:\ >> #{output_file}
      dir /s "c:\Documents and Settings" >> #{output_file}
      dir /s "c:\Program Files\" >> #{output_file}
      dir "%systemdrive%\Users\*.*" >> #{output_file}
      dir "%userprofile%\AppData\Roaming\Microsoft\Windows\Recent\*.*" >> #{output_file}
      dir "%userprofile%\Desktop\*.*" >> #{output_file}
      tree /F >> #{output_file}
    cleanup_command: |
      del #{output_file}
    name: command_prompt
- name: File and Directory Discovery (PowerShell)
```

```
      name: command_prompt
      elevation_required: false
citadmin@citadmin-VirtualBox:~/atomic-red-team/atomics/T1083$ find /home -type f -name "*.txt"
find: ‘/home/user03’: Permission denied
find: ‘/home/student’: Permission denied
find: ‘/home/user02’: Permission denied
/home/citadmin/atomic-red-team/atomics/T1053.005/src/T1053.005-macrocode.txt
/home/citadmin/atomic-red-team/atomics/T1003.003/src/diskshadow.txt
/home/citadmin/atomic-red-team/atomics/T1133/src/list of vpn extension.txt
/home/citadmin/atomic-red-team/atomics/T1016/src/top-128.txt
/home/citadmin/atomic-red-team/atomics/T1204.002/src/chromeexec-macrocode.txt
/home/citadmin/atomic-red-team/atomics/T1204.002/src/test9-GenericPayloadDownload.txt
/home/citadmin/atomic-red-team/atomics/T1204.002/src/Mirror Blast.txt
/home/citadmin/atomic-red-team/atomics/T1204.002/src/test9-example-payload.txt
/home/citadmin/atomic-red-team/atomics/T1204.002/src/payload.txt
/home/citadmin/atomic-red-team/atomics/T1059.005/src/T1059.005-macrocode.txt
/home/citadmin/atomic-red-team/atomics/T1059.005/src/T1059_005-macrocode.txt
/home/citadmin/atomic-red-team/atomics/T1218.008/src/odbcconf net.40/packages/DllExport.1.5.2/License.txt
/home/citadmin/atomic-red-team/atomics/T1218.008/src/odbcconf net.40/packages/DllExport.1.5.2/changelog.txt
/home/citadmin/atomic-red-team/atomics/T1218.008/src/odbcconf net.40/packages/DllExport.1.5.2/tools/coreclr/changelog.txt
/home/citadmin/atomic-red-team/atomics/T1218.008/src/odbcconf net.40/packages/DllExport.1.5.2/tools/coreclr/_Version.txt
/home/citadmin/atomic-red-team/atomics/T1218.008/src/odbcconf net.40/packages/DllExport.1.5.2/3rd-party.txt
/home/citadmin/atomic-red-team/atomics/T1055/src/x64/T1055-macrocode.txt
/home/citadmin/atomic-red-team/atomics/T1595.003/src/wordlist.txt
/home/citadmin/atomic-red-team/atomics/T1221/src/readme.txt
/home/citadmin/atomic-red-team/atomics/T1110.001/src/passwords.txt
/home/citadmin/atomic-red-team/atomics/T1110.001/src/bruteuser.txt
/home/citadmin/atomic-red-team/atomics/T1129/src/esxi_vibremove.txt
/home/citadmin/atomic-red-team/atomics/T1129/src/esxi_vibinstall.txt
/home/citadmin/atomic-red-team/atomics/used_guids.txt
/home/citadmin/atomic-red-team/atomics/T1564/src/T1564-macrocode.txt
/home/citadmin/atomic-red-team/atomics/T1529/src/esx_avoslocker_kill_vm.txt
/home/citadmin/atomic-red-team/atomics/T1529/src/esx_pkill.txt
```

After accessing the atomic red team technique T1083 we used the ***“cat”*** command to display the contents of the **T1083.yaml** files after that we scanned for text file in the user’s home directory. This simulated an attacker scanning the file system for readable .txt files, which could include configuration files, saved passwords, or notes. Allowing broad read-access to sensitive .txt files can expose credentials, internal documentation, or system details. Monitoring file access activities or applying “***chmod***” permissions could help mitigate the risks of this attack. For example I accessed the passwords.txt which is usually used for brute-Force it contains common passwords that could be applied to our system to

gain access:



- **T1016-System Network Configuration Discovery.**

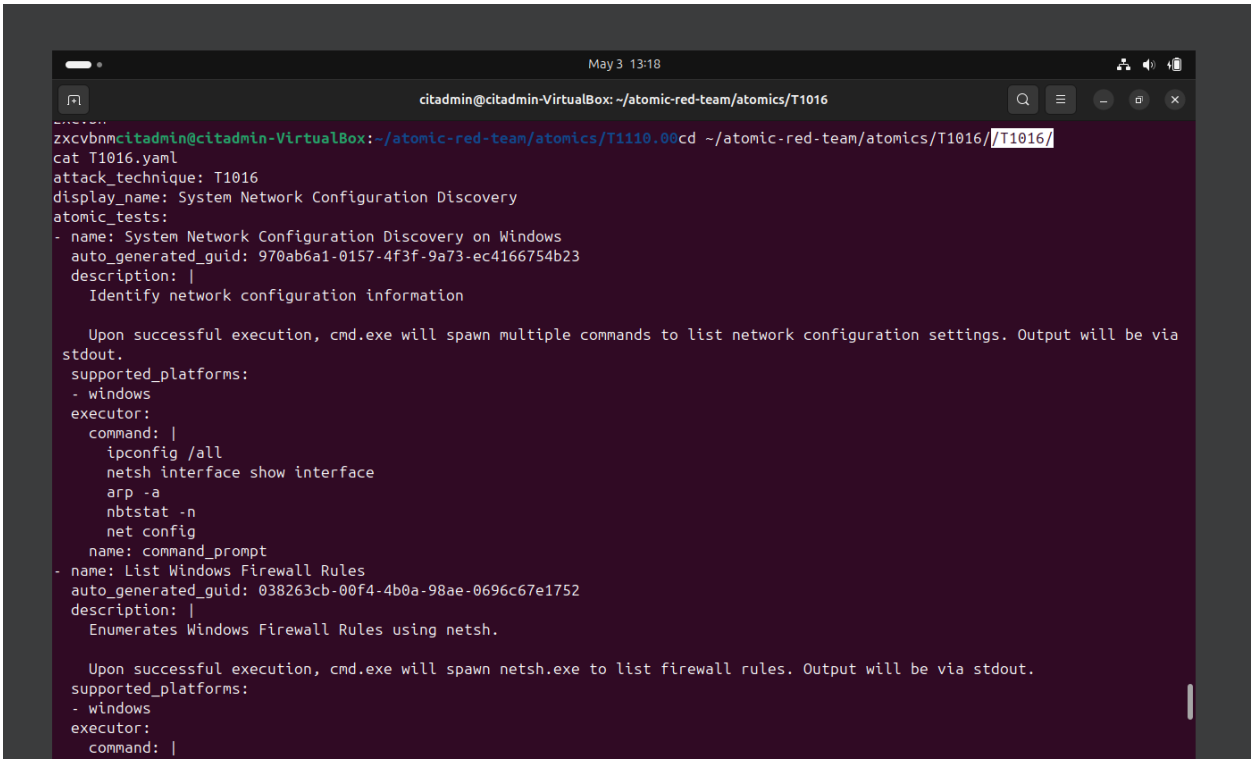
This attack simulates what a malicious actor would do to list network information on a compromised machine. This includes:

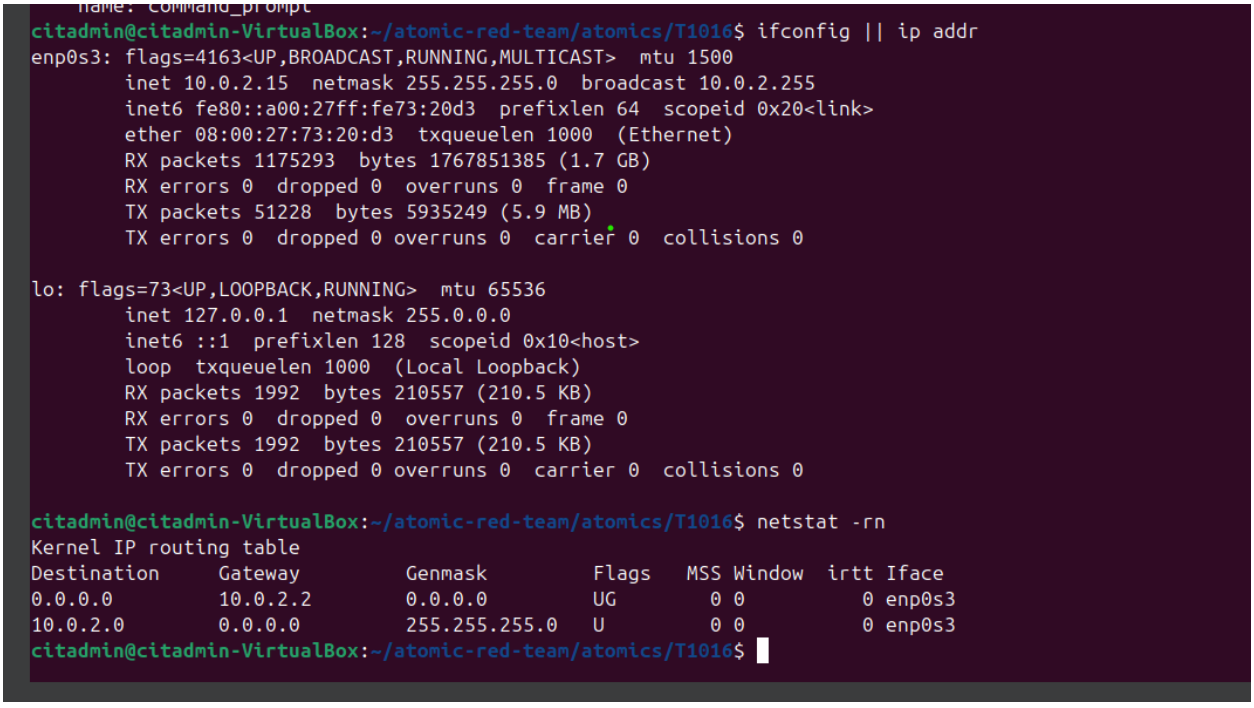
- IP addresses
- Routing tables
- MAC addresses
- Active interfaces
- Listening ports

This type of information allows attackers to:

- Identify target systems in the network
- Understand firewall protections

- Map out the internal network for lateral movement
- Pinpoint vulnerable services





After displaying the contents of attack T1016, The YAML file revealed the use of commands

such as arp, ip addr, and netstat, (for linux systems) which attackers use to enumerate local IPs, interfaces, and open network connections.

We then used the ***ifconfig || ip addr*** (which ran either one of the command while making sure whichever is installed runs) command to display Network interfaces and IP addresses, MACs.
We also ran netstat -rn which shoes the Kernel routing table, it displayed our system gateway, network routes and interface bindings. These information are critical for attackers as it helps them map out the system network layout and identify reachable hosts, these actions aren't usually flagged by detection tools.

- **T1059- Command and Scripting Interpreter**

```
citadmin@citadmin-VirtualBox:~/atomic-red-team/atomics/T1016$ cd ~/atomic-red-team/atomics/T1059/
citadmin@citadmin-VirtualBox:~/atomic-red-team/atomics/T1059$ cat T1059.yaml
attack_technique: T1059
display_name: 'Command and Scripting Interpreter'
atomic_tests:
- name: AutoIt Script Execution
  auto_generated_guid: a9b93f17-31cb-435d-a462-5e838a2a6026
  description: |
    An adversary may attempt to execute suspicious or malicious script using AutoIt software instead of regular terminal like
powershell or cmd. Calculator will popup when the script is executed successfully.
  supported_platforms:
  - windows
  dependency_executor_name: powershell
  dependencies:
  - description: |
      AutoIt executable file must exist on disk at the specified location (#{autoit_path})
    prereq_command: |
      if(Test-Path "#{autoit_path}") {
          exit 0
      } else {
          exit 1
      }
    get_prereq_command: |
      New-Item -Type Directory "PathToAtomicsFolder\..\ExternalPayloads\" -ErrorAction Ignore -Force | Out-Null
      $AutoItURL = "https://www.autoitscript.com/cgi-bin/getfile.pl?autoit3/autoit-v3-setup.exe"
      $InstallerPath = "$PathToAtomicsFolder\..\ExternalPayloads\autoit-v3-setup.exe"
      Invoke-WebRequest -Uri $AutoItURL -OutFile $InstallerPath
      Start-Process -FilePath $InstallerPath -ArgumentList "/S" -Wait
```

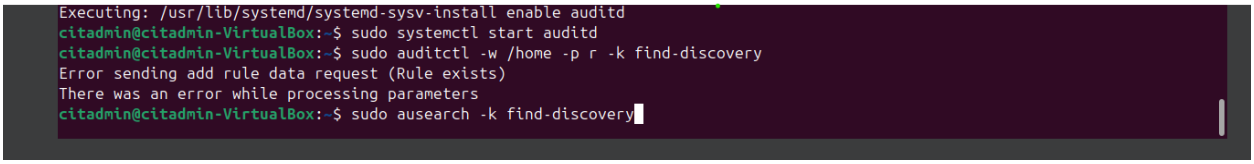
```
    name: powershell
citadmin@citadmin-VirtualBox:~/atomic-red-team/atomics/T1059$ whoami && id && uname -a
citadmin
uid=1000(citadmin) gid=1000(citadmin) groups=1000(citadmin),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),114(lpadm
in)
Linux citadmin-VirtualBox 6.8.0-52-generic #53-Ubuntu SMP PREEMPT_DYNAMIC Sat Jan 11 00:06:25 UTC 2025 x86_64 x86_64 x86_64 GN
U/Linux
```

This attack, simulates an attacker executing system-level commands through a shell after gaining access. The ***whoami*** and *id* commands confirmed that the attacker had access to a user with sudo (admin) privileges. ***uname -a*** revealed the OS and kernel, which is useful for targeting the system with known exploits.

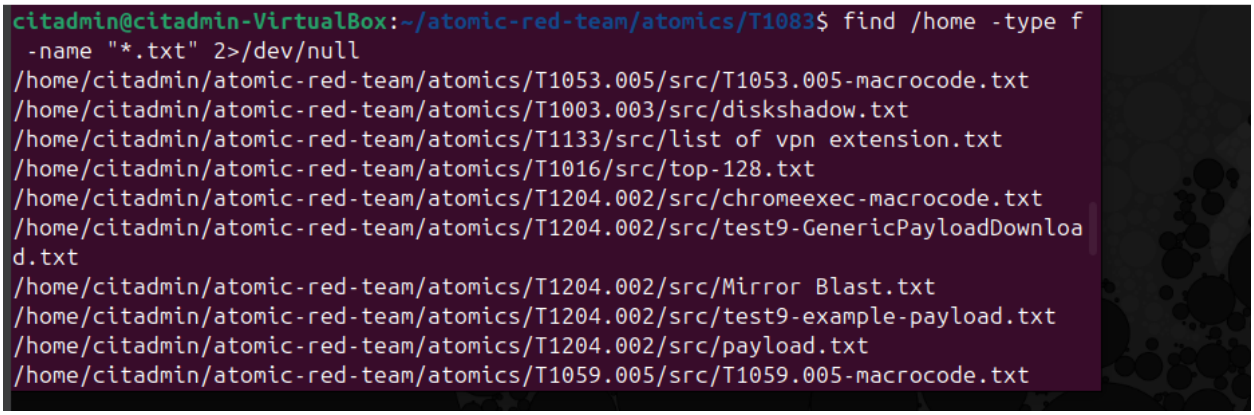
Task 4 Detection and Monitoring

- **T1083 – File and Directory Discovery**

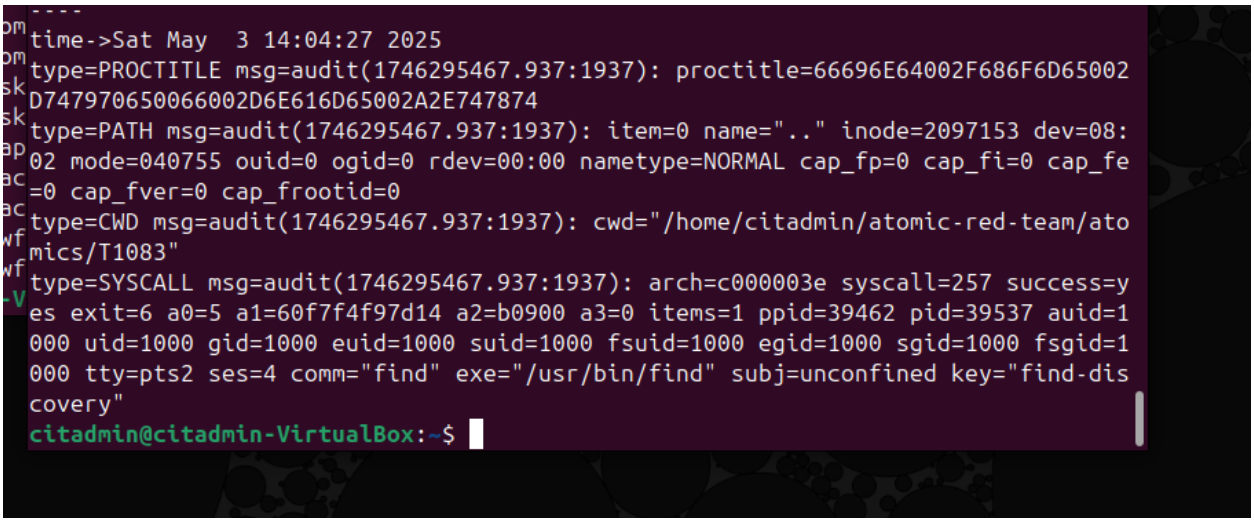
For this one I'll be installing ***auditd*** which stands for stands for Audit Daemon. It's a Linux service that provides a powerful way to track and log system activity in real-time, especially: File access (read/write), Command execution, Privilege changes, System call usage, User login and logout activity



After installing auditd we've enabled it and set a rule to the /home directory so that whenever it is accessed it's flagged and tagged with "find discovery"



Here we accessed the /home folder to see if it'll be flagged



Here's proof that after accessing the folder, it is flagged by auditd.

- **T1016- System Network Configuration**

For this one we'll also use auditd and add rules to monitor network tools and when they are used, here we'll change our rule and how it will work:
sudo auditctl -w <file_path> -p <permissions> -k <custom_key>

Our 3 new rules:



Running ifconfig,netstat and ip addr:

```
citadmin@citadmin-VirtualBox:~/atomic-red-team/atomics/T1016$ cd
citadmin@citadmin-VirtualBox:~$ ifconfig || ip addr
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe73:20d3  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:73:20:d3  txqueuelen 1000  (Ethernet)
        RX packets 1177668  bytes 1769107044 (1.7 GB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 53009  bytes 7142083 (7.1 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 2491  bytes 266679 (266.6 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2491  bytes 266679 (266.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

citadmin@citadmin-VirtualBox:~$ netstat -rn
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         10.0.2.2        0.0.0.0         UG        0 0          0 enp0s3
10.0.2.0        0.0.0.0         255.255.255.0   U         0 0          0 enp0s3
citadmin@citadmin-VirtualBox:~$
```

Using audit to check if they are recorded:

```
citadmin@citadmin-VirtualBox:~$ sudo auditctl -w /usr/sbin/ip -p x -k net-recon
citadmin@citadmin-VirtualBox:~$ sudo ausearch -k net-recon
----
time->Sat May  3 14:21:07 2025
type=PROCTITLE msg=audit(1746296467.009:2168): proctitle=6175646974637 46C002D77002F7573722F7362696E2F6966636F6E666967002D700078002D6B006E65742D7265636F6E
type=PATH msg=audit(1746296467.009:2168): item=0 name="/usr/sbin/" inode=393225 dev=08:02 mode=040755 ouid=0 ogid=0 rdev=00:00 nametype=PARENT cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1746296467.009:2168): cwd="/home/citadmin"
type=SOCKADDR msg=audit(1746296467.009:2168): saddr=100000000000000000000000
type=SYSCALL msg=audit(1746296467.009:2168): arch=c000003e syscall=44 success=yes exit=1084 a0=4 a1=7ffeac0474d0 a2=43c a3=0 items=1 ppid=40206 pid=40207 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts2 ses=4 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1746296467.009:2168): auid=1000 ses=4 subj=unconfined op=add_rule key="net-recon" list=4 res=1
----
time->Sat May  3 14:21:15 2025
type=PROCTITLE msg=audit(1746296475.671:2175): proctitle=6175646974637 46C002D77002F7573722F62696E2F6E657473746174002D700078002D6B006E65742D7265636F6E
type=PATH msg=audit(1746296475.671:2175): item=0 name="/usr/bin/" inode=393218 dev=08:02 mode=040755 ouid=0 ogid=0 rdev=00:00 nametype=PARENT cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1746296475.671:2175): cwd="/home/citadmin"
type=SOCKADDR msg=audit(1746296475.671:2175): saddr=100000000000000000000000
type=SYSCALL msg=audit(1746296475.671:2175): arch=c000003e syscall=44 success=yes exit=1084 a0=4 a1=7fffaf7ffd50 a2=43c a3=0 items=1 ppid=40216 pid=40217 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts2 ses=4 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1746296475.671:2175): auid=1000 ses=4 subj=unconfined op=add_rule key="net-recon" list=4 res=1
----
time->Sat May  3 14:21:24 2025
type=PROCTITLE msg=audit(1746296484.314:2182): proctitle=6175646974637 46C002D77002F7573722F7362696E2F6970002D700078002D6B006E65742D7265636F6E
type=PATH msg=audit(1746296484.314:2182): item=0 name="/usr/sbin/" inode=393225 dev=08:02 mode=040755 ouid=0 ogid=0 rdev=00:00
```

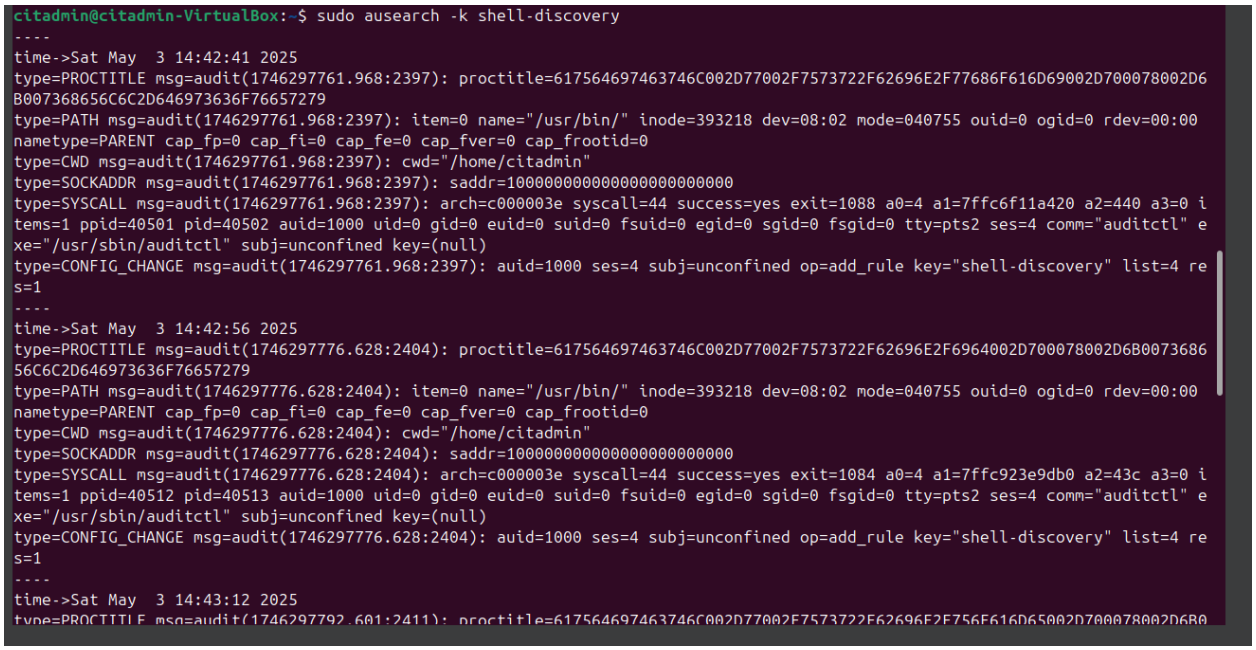
```
----
time->Sat May  3 14:22:43 2025
type=PROCTITLE msg=audit(1746296563.084:2185): proctitle="ifconfig"
type=PATH msg=audit(1746296563.084:2185): item=1 name="/lib64/ld-linux-x86-64.so.2" inode=419159 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1746296563.084:2185): item=0 name="/usr/sbin/ifconfig" inode=431175 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1746296563.084:2185): cwd="/home/citadmin"
type=EXECVE msg=audit(1746296563.084:2185): argc=1 a0="ifconfig"
type=SYSCALL msg=audit(1746296563.084:2185): arch=c000003e syscall=59 success=yes exit=0 a0=5a76653e9000 a1=5a76653e8ea0 a2=5a76653e4110 a3=5a76653e8ea0 items=2 ppid=4641 pid=40240 auid=1000 uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000 sgid=1000 fsgid=1000 tty=pts0 ses=4 comm="ifconfig" exe="/usr/sbin/ifconfig" subj=unconfined key="net-recon"
----
time->Sat May  3 14:22:54 2025
type=PROCTITLE msg=audit(1746296574.752:2186): proctitle=6E6574737461740 02D726E
type=PATH msg=audit(1746296574.752:2186): item=1 name="/lib64/ld-linux-x86-64.so.2" inode=419159 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1746296574.752:2186): item=0 name="/usr/bin/netstat" inode=431174 dev=08:02 mode=0100755 ouid=0 ogid=0 rde
```

- **T1059 – Command and Scripting Interpreter.**

We added 3 new rule to monitor ***whoami, id*** and ***uname*** commands

```
egid=1000 sgid=1000 fsgid=1000 tty=pts0 ses=4 comm="netstat" exe="/usr/bin/netstat" subj=unconfined key="net-recon"
citadmin@citadmin-VirtualBox:~$ sudo auditctl -w /usr/bin/whoami -p x -k shell-discovery
[sudo] password for citadmin:
citadmin@citadmin-VirtualBox:~$ sudo auditctl -w /usr/bin/id -p x -k shell-discovery
citadmin@citadmin-VirtualBox:~$ sudo auditctl -w /usr/bin/uname -p x -k shell-discovery
citadmin@citadmin-VirtualBox:~$
```

Running whoami,id and uname commands to check if they'll be flagged/logged.



In all, ***auditd*** is a very powerful and flexible tool capable of detecting any attack activity in the system if a rule is implemented on it for the specific attack. It doesn't require any third party software and would be extremely useful for incident responses

Task 5: Incident Response Report

Simulated activity of MITRE ATT&CK tactics T1083 (File and Directory Discovery), T1016 (System Network Configuration Discovery), and T1059 (Command and Scripting Interpreter) were carried out effectively utilizing Atomic Red Team on the identified Linux virtual machine. Subsequent simulated activities were then picked up through monitoring capabilities of the auditd service, which had specifically been configured with rules to check for such actions.

The identification of these staged events relied directly on the custom auditd rules implemented in the previous phase. Specifically, attempts to scan the /home directory based on T1083 triggered warnings tagged with the "find_discovery" key, marking unauthorized file system discovery. Similarly, the calling of network command information such as ifconfig, ip addr, and netstat -rn, characteristic of T1016, invoked rules monitoring

their respective binary addresses (e.g., /sbin/ifconfig, /bin/netstat) and logging with the "network_tool_used" key. Finally, calling commands such as whoami, id, and uname -a for system discovery (T1059) invoked rules monitoring said individual binaries, resulting in logs tagged with the "user_info_cmd" key. The auditd logs provided definite evidence that these pre-defined suspicious behaviors had occurred.

In being caught via auditd alerts, the first containment strategy in a real-world environment would be to isolate the compromised Linux VM from the network. This could be accomplished by shutting down its virtual network interface or implementing strict firewall policies, thereby halting potential lateral movement or communication with external command-and-control servers. At the same time, the auditd logs would be examined to determine the specific process ID (PID) and user ID (UID) responsible for the reported activities. Based on this, the suspicious processes would be terminated (e.g., kill) and, if a user account was found compromised, locked (passwd -l) or its active sessions forced to terminate (pkill -u) to prevent further malicious activity.

Eradication in this case focuses on eliminating the adversary's ability to perform these activities, since the emulated TTPs were executing legitimate system commands rather than installing long-term malware. This involves a thorough review of system and audit logs (ausearch, journalctl, /var/log/auth.log) and user shell histories (.bash_history) to ascertain the full scope of the adversary's activity and ensure all unauthorized activity is understood. Confirmation of escalated privilege levels used at the time of execution (e.g., use of sudo) is also critical in gaining insight into the depth of compromise and verifying how elevated privileges have been achieved or misused. Removal of incidents focuses here on removing unwanted access or capability as opposed to being required to remove files.

Recovery processes include bringing the system to a known good and secure operational state. After the threat has been contained and neutralized, the isolated VM would have network connectivity restored. Any user credentials suspected to have been compromised during the incident would be reset. Verification of critical system files and settings would be done to ensure system integrity. Most significantly, the recovery phase directly follows up with active hardening, using the lessons learned during the incident in order to prevent recurrence, as described in the System Hardening task.

The exercise demonstrated that auditd is an effective utility for the detection of specific, predefined malicious behavior, effectively labeling the simulated TTPs based on the given rules. But its value is entirely dependent on the quality and scope of those rules; activities not directly monitored would be out of sight. Moreover, although auditd exports raw logging data, establishing intent and correlating many disparate events typically requires

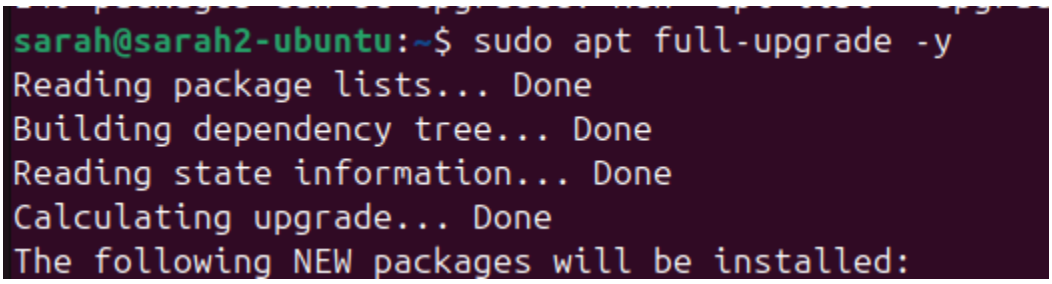
additional examination or use within more advanced security tools like a SIEM. This simulation highlighted the need to track execution of shared, native system utilities because attackers constantly use them throughout the reconnaissance and execution phases of a penetration.

Task 6: System Hardening

This report summarizes the system hardening measures implemented on the Linux virtual machine as part of Task 6 of the project. Once the adversary emulation exercises had been conducted using Atomic Red Team and detection analysis subsequently performed, the following measures were undertaken to mitigate identified risks, reduce the overall attack surface, and enhance the security posture of the environment in line with security best practices. The actions performed and their outcomes are documented below, with the assistance of screenshots captured during the process.

System Patching and Updates

The first thing was to make sure the system was current with security updates. Package repositories were refreshed by running sudo apt update, and then all packages installed on the system were updated to their latest versions using sudo apt upgrade -y. This process resolves known vulnerabilities that have been addressed by software vendors since the system was last updated. It's essential to install these patches as early as possible since it completes reported security holes which could otherwise be used by the attackers. .



```
sarah@sarah2-ubuntu:~$ sudo apt autoremove -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
```

Firewall Configuration

The network perimeter was secured by configuring the host-based Uncomplicated Firewall. UFW was enabled with the policies set to block all incoming traffic by default but allow outgoing connections. Rules were specifically configured to permit traffic only for necessary services, primarily to allow incoming connections on TCP port 22 for SSH access. This configuration substantially reduces the exposed network attack surface, rendering external network scanning (in relation to T1016) less useful and preventing unauthorized connections to potentially vulnerable services.

```
sarah@sarah2-ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

```
sarah@sarah2-ubuntu:~$ sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     ALLOW IN    Anywhere
[ 2] 22/tcp (v6)                ALLOW IN    Anywhere (v6)
```

File Permission Review and Hardening

Security within the file system was addressed through the re-examining and correcting of permissions. Files suspected of being potentially sensitive in the past, i.e., /var/cache/debconf/passwords.dat, were checked via ls -l on permissions. Where permissions had been too lenient, the principle of least privilege was executed by running sudo chmod to grant generally only read/write access via 600 to the owner. This action directly removes the danger of sensitive data leakage through File and Directory Discovery techniques (T1083). Files that were determined to be non-sensitive documents, like AWS

CLI examples, were logged but did not require permissions updated

```
sarah@sarah2-ubuntu:~$ sudo find /etc /home /var /opt -type f -name "*password*'

sudo find /etc /home /var /opt -type f -name "*secret*"
sudo find /etc /home /var /opt -type f -name "*credential*"
sudo find /etc /home /var /opt -type f -name "*backup*"
sudo find /etc /home /var /opt -type f -name "*.bak"
```

```
sarah@sarah2-ubuntu:~$ ls -l /var/cache/debconf/passwords.dat # Should now show
-rw-------
ls -l /var/lib/cloud/instances/iid-datasource-none/sem/config_set_passwords # Sh
ould now show -rw-------
-rw------- 1 root root 0 Feb 15 08:09 /var/cache/debconf/passwords.dat
-rw------- 1 root root 25 Feb 26 00:16 /var/lib/cloud/instances/iid-datasource-n
one/sem/config_set_passwords
```

Disabling Unncecessary Services

To better minimize the attack surface, running services were examined with sudo ss -tulnp. Services discovered to be unnecessary for the VM to function , that is, the nginx web server, the cupsd print service, and the avahi-daemon zeroconf network service, were stopped with sudo systemctl stop and disallowed from auto-start on boot with sudo systemctl disable. The deactivation of the services must be confirmed through their status reports. The following reduction in listening network ports should be evident in the terminal ss output. Shutting down the services removes potential attack vectors and reduces information leak beneficial for reconnaissance (T1016). Docker services were left running but not disabled; disabling docker.service would be the process if Docker is found to be unnecessary.

```
sarah@sarah2-ubuntu:~$ sudo systemctl stop nginx.service
sudo systemctl disable nginx.service
sudo systemctl status nginx.service
Synchronizing state of nginx.service with SysV service script with /usr/lib/syst
emd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable nginx
Removed "/etc/systemd/system/multi-user.target.wants/nginx.service".
○ nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: e>
     Active: inactive (dead)
       Docs: man:nginx(8)
```

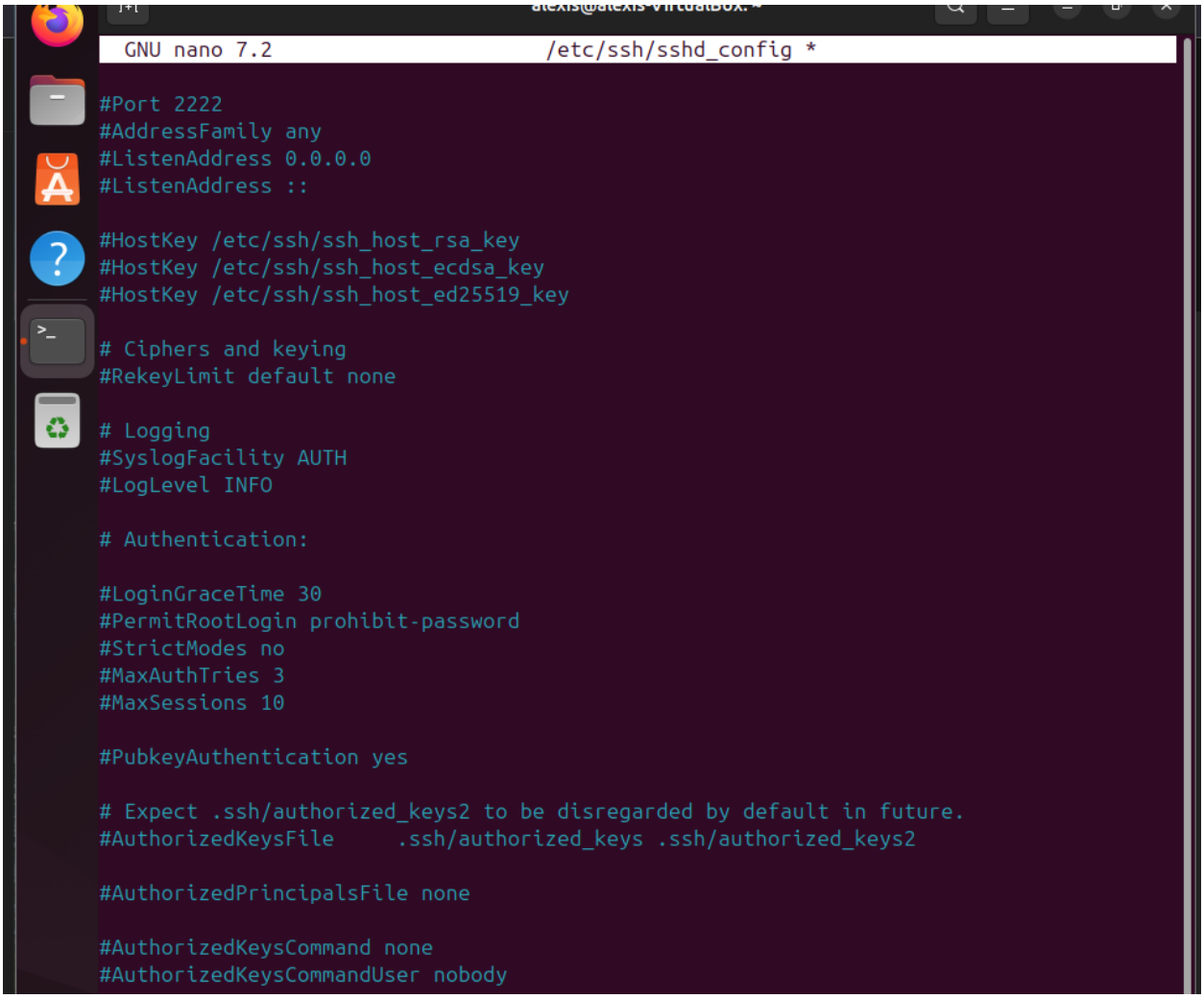
```
sarah@sarah2-ubuntu:~$ sudo systemctl stop cups.service
sarah@sarah2-ubuntu:~$ sudo systemctl disable cups.service
sudo systemctl disable cups.socket
sudo systemctl disable cups.path
Synchronizing state of cups.service with SysV service script with /usr/lib/syste
md/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable cups
Removed "/etc/systemd/system/printer.target.wants/cups.service".
Removed "/etc/systemd/system/multi-user.target.wants/cups.service".
Removed "/etc/systemd/system/multi-user.target.wants/cups.path".
Removed "/etc/systemd/system/sockets.target.wants/cups.socket".
sarah@sarah2-ubuntu:~$ sudo systemctl status cups.service
○ cups.service - CUPS Scheduler
     Loaded: loaded (/usr/lib/systemd/system/cups.service; disabled; preset: en>
     Active: inactive (dead) since Sun 2025-05-04 18:06:41 UTC; 5min ago
   Duration: 40min 14.251s
TriggeredBy: ○ cups.socket
       Docs: man:cupsd(8)
   Main PID: 19367 (code=exited, status=0/SUCCESS)
     Status: "Scheduler is running..."
        CPU: 61ms
```

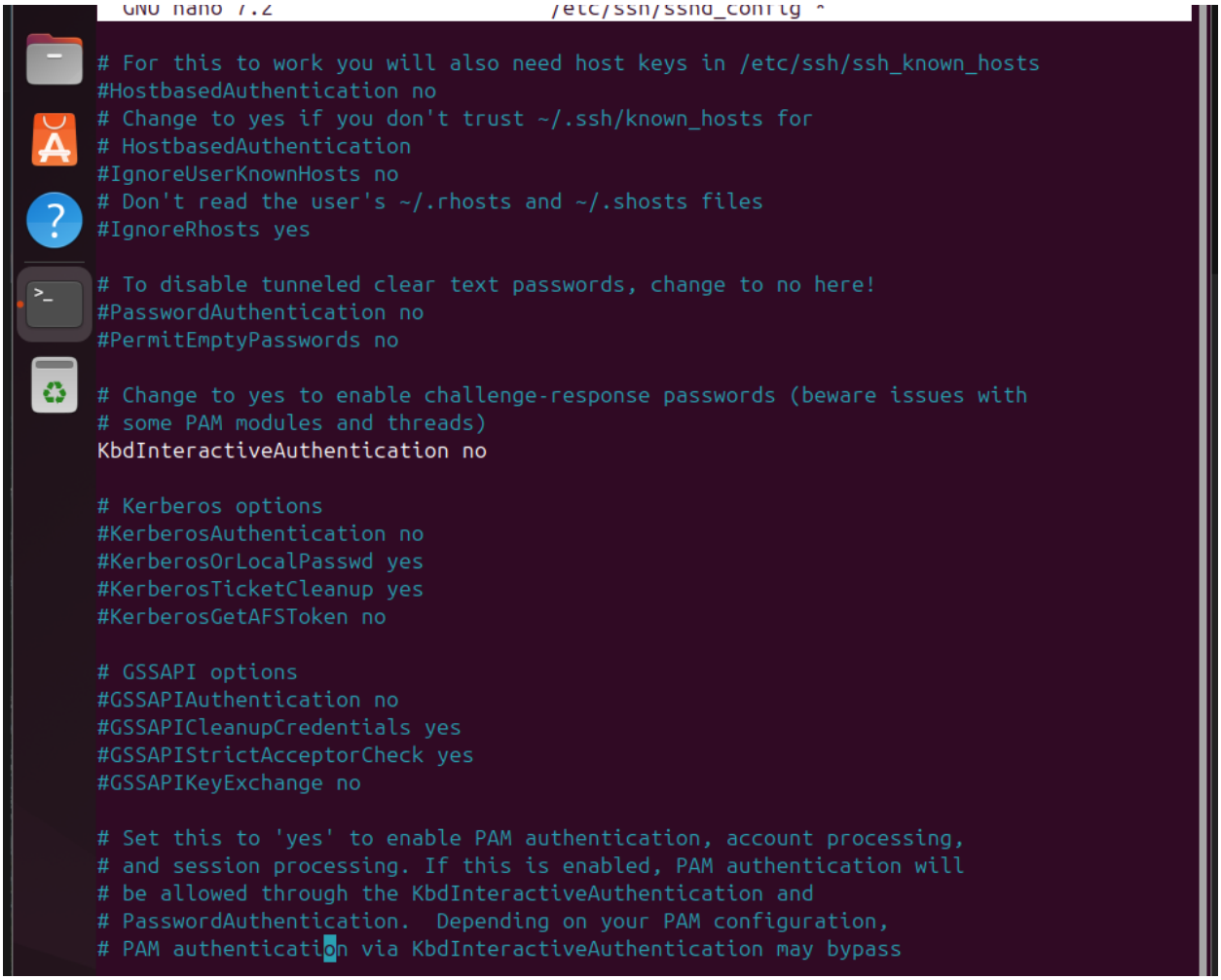
```
sarah@sarah2-ubuntu:~$ sudo systemctl stop avahi-daemon.service
Stopping 'avahi-daemon.service', but its triggering units are still active:
avahi-daemon.socket
sarah@sarah2-ubuntu:~$ sudo systemctl disable avahi-daemon.service
sudo systemctl disable avahi-daemon.socket
Removed "/etc/systemd/system/dbus-org.freedesktop.Avahi.service".
Removed "/etc/systemd/system/multi-user.target.wants/avahi-daemon.service".
Removed "/etc/systemd/system/sockets.target.wants/avahi-daemon.socket".
Disabling 'avahi-daemon.service', but its triggering units are still active:
avahi-daemon.socket
sarah@sarah2-ubuntu:~$ sudo systemctl status avahi-daemon.service
○ avahi-daemon.service - Avahi mDNS/DNS-SD Stack
     Loaded: loaded (/usr/lib/systemd/system/avahi-daemon.service; disabled; pr>
     Active: inactive (dead) since Sun 2025-05-04 18:14:05 UTC; 34s ago
   Duration: 2d 1h 30min 9.703s
TriggeredBy: ● avahi-daemon.socket
   Main PID: 858 (code=exited, status=0/SUCCESS)
     Status: "avahi-daemon 0.8 starting up."
        CPU: 504ms
```

```
sarah@sarah2-ubuntu:~$ sudo ss -tulnp
Netid    State     Recv-Q    Send-Q        Local Address:Port       Peer Address:Port
Process
udp      UNCONN    0         0                 0.0.0.0:5353            0.0.0.0:*
 users:(("avahi-daemon",pid=858,fd=12))
udp      UNCONN    0         0                 0.0.0.0:49102           0.0.0.0:*
 users:(("avahi-daemon",pid=858,fd=14))
udp      UNCONN    0         0              127.0.0.54:53              0.0.0.0:*
 users:(("systemd-resolve",pid=30315,fd=16))
udp      UNCONN    0         0           127.0.0.53%lo:53              0.0.0.0:*
 users:(("systemd-resolve",pid=30315,fd=14))
udp      UNCONN    0         0                    [::]:5353               [::]:*
 users:(("avahi-daemon",pid=858,fd=13))
udp      UNCONN    0         0                    [::]:57478              [::]:*
 users:(("avahi-daemon",pid=858,fd=15))
tcp      LISTEN    0         4096           127.0.0.1:631             0.0.0.0:*
 users:(("cupsd",pid=19367,fd=7))
tcp      LISTEN    0         4096        127.0.0.53%lo:53             0.0.0.0:*
 users:(("systemd-resolve",pid=30315,fd=15))
tcp      LISTEN    0         4096              0.0.0.0:4510           0.0.0.0:*
 users:(("docker-proxy",pid=28210,fd=7))
tcp      LISTEN    0         4096              0.0.0.0:4511           0.0.0.0:*
 users:(("docker-proxy",pid=28225,fd=7))
tcp      LISTEN    0         4096              0.0.0.0:4526           0.0.0.0:*
 users:(("docker-proxy",pid=28480,fd=7))
tcp      LISTEN    0         4096              0.0.0.0:4527           0.0.0.0:*
 users:(("docker-proxy",pid=28498,fd=7))
tcp      LISTEN    0         4096              0.0.0.0:4524           0.0.0.0:*
 users:(("docker-proxy",pid=28445,fd=7))
tcp      LISTEN    0         4096              0.0.0.0:4525           0.0.0.0:*
 users:(("docker-proxy",pid=28464,fd=7))
tcp      LISTEN    0         4096              0.0.0.0:4522           0.0.0.0:*
 users:(("docker-proxy",pid=28415,fd=7))
tcp      LISTEN    0         4096              0.0.0.0:4523           0.0.0.0:*
 users:(("docker-proxy",pid=28431,fd=7))
tcp      LISTEN    0         4096              0.0.0.0:4520           0.0.0.0:*
 users:(("docker-proxy",pid=28378,fd=7))
tcp      LISTEN    0         4096           127.0.0.54:53             0.0.0.0:*
 users:(("systemd-resolve",pid=30315,fd=17))
```

Trash

SSH Hardening:


GNU nano 7.2 /etc/ssh/sshd_config *
#Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key
Ciphers and keying
#RekeyLimit default none
Logging
#SyslogFacility AUTH
#LogLevel INFO
Authentication:
#LoginGraceTime 30
#PermitRootLogin prohibit-password
#StrictModes no
#MaxAuthTries 3
#MaxSessions 10
#PubkeyAuthentication yes
Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
#AuthorizedPrincipalsFile none
#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody



To enhance the security of the SSH service on the system, several key Configurations were implemented such as:

- Port 2222
  - Changed the default SSH port from 22 to 2222 to reduce exposure to automated scanning and brute force attempts
- PermitRootLogin no:
  - Disabled direct root login over SSH to prevent attackers from targeting the root account directly
- PasswordAuthentication no:
  - Disabled password-based authentication, requiring more secure methods such as SSH keys for login
- MaxAuthTries 3:
  - Limited the number of allowed attempts per connection to 3
- LoginGraceTime 30:

- Reduced the login grace time to 30 seconds, minimimizing the windows for unauthorized login attempts

Conclusion

The hardening controls implemented as specified in this report, various from patching, firewall configuration, file permission change, service reduction, user privilege analysis, SSH security enhancement, and audit system validation, have all converged to develop a more secure position for the Linux virtual machine. These multi-layered defenses oppose threats associated with the simulated adversary techniques (T1083, T1016, T1059) using security best practices in general, offering an attack-reduced surface area and increased resilience against likely threats. Automated maintenance and continuous monitoring need to be applied so that the heightened state of security is sustained.